

RelateOS (脈絡關係)

Full Technical Specification (PRD)

Engineering-Grade Blueprint for MVP Implementation

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Abstract

This document is the complete, self-contained, engineering-grade Product Requirements Document (PRD) for RelateOS. It has been refined and expanded to include **every single detail** required for a development team to build, test, deploy, and maintain the MVP without further clarification. All architectural decisions, data models, API contracts, edge cases, security rules, performance budgets, UI flows, native bridge implementations, AI prompt templates, and compliance requirements are explicitly defined. The specification resolves the previous iOS keyboard memory constraint by mandating a secure main-app + Cloudflare proxy architecture.

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Product Vision & Scope

RelateOS is a context-aware relationship intelligence platform purpose-built for the Hong Kong market. It functions as a system-level assistant that decodes multilingual (Cantonese/English/-Mandarin) communication nuances and actively mitigates “face-saving” conflict escalation in high-context Asian relationships.

1.1 In-Scope for MVP (Weeks 1–10)

- iOS Custom Keyboard Extension (UIInputViewController) + Android InputMethodService.
- On-device text capture of the last 20 messages, securely bridged to a Cloudflare Workers proxy.
- Solo-Mode Relationship Health Scoring engine with rolling 7-day window.
- Wisdom Circle: anonymous community forum with automated PII redaction via Edge Function.
- Supabase Auth (email + Apple Sign-In + Google Sign-In) with Row-Level Security.
- Flutter-based main app for onboarding, dashboard, and history.

1.2 Out-of-Scope for MVP (Deferred)

- Real-time Voice Note transcription (deferred to v1.2).
- Direct Octopus/MTR API integration (MVP uses OCR screenshot workaround via MLKit on-device).
- Shared Couples Dashboard (deferred to v1.5).
- Proactive push notifications based on score drops (deferred to v1.1).
- Multi-user relationship linking.

System Architecture & Tech Stack

The architecture is hybrid to respect iOS keyboard extension memory limits (< 50 MB) and Android background constraints while delivering sub-second latency.

2.1 Frontend

- Flutter 3.24+ (stable channel as of April 2026).
- State management: `flutter_riverpod` 2.5+ (Provider + Notifier pattern).
- UI: Material 3 + Cupertino for platform fidelity.
- Packages (exact versions locked in `pubspec.yaml`):
 - `supabase_flutter`: `^2.8.0`
 - `flutter_secure_storage`: `^9.2.0`
 - `http`: `^1.2.0` (for direct Cloudflare calls when needed)
 - `flutter_localizations` + `intl` for zh-HK / en / zh-CN

2.2 Native Bridges (Platform Channels)

- **iOS**: Swift 6.0 Custom Keyboard Extension (`UIInputViewController` subclass). Uses `AppGroups` (`group.com.relateos.keyboard`) to share `UserDefaults` and `Keychain` with main app. Text capture via `textDocumentProxy.documentContextBeforeInput` + `selectedText`.
- **Android**: Kotlin `InputMethodService`. Uses `InputConnection` to read `getTextBeforeCursor(200, 0)` and `getTextAfterCursor(200, 0)`. Accessibility fallback for non-editable fields.
- Method channel name: `com.relateos/keyboard_bridge`.

2.3 Backend & Data

- Supabase (PostgreSQL 15, Edge Functions, Realtime, Storage).
- Auth: Supabase Auth with RLS policies.
- AI Routing: Cloudflare Workers (global edge, rate limiting at 60 req/min per user).
- Tier-2 AI: Google Gemini 2.5 Pro (via official REST API through Cloudflare proxy).

2.4 AI Pipeline (Mandatory Flow)

Every keyboard request MUST follow:

1. Keyboard → Platform Channel → Main Flutter app (background isolate).
2. Main app encrypts payload (AES-256-GCM, key derived from Supabase JWT).
3. POST to Cloudflare Worker (<https://relateos-ai-proxy.relateos.workers.dev>).
4. Worker performs PII scrub + rate limit.
5. Worker forwards to Gemini 2.5 Pro with strict system prompt (see Section 3.2).
6. Response decrypted and returned to keyboard UI in < 1200 ms.

Detailed Core Features & Epics

3.1 Epic 0: Onboarding & Authentication

1. Splash → Apple/Google/Email sign-up (Supabase).
2. Language selection (default: zh-HK).
3. Explicit consent modal for keyboard full-access (iOS) / Accessibility + Draw-over (Android). Event logged to `users.consent_keyboard_granted` (TIMESTAMPZ, required).
4. Baseline interaction style quiz (5 questions) to initialise dynamic weights w_1, w_2, w_3 .
5. Dashboard with first Health Score = 0.0.

3.2 Epic 1: Zero-Friction Engine (Keyboard Extension)

Trigger: User focuses any text field in target apps (WhatsApp, WeChat, IG, Telegram, Signal, Messages).

iOS Implementation Details:

- Extension size limit enforced: < 40 MB compiled.
- Use `AppGroups` suite for sharing last 20 messages.
- UI: 3 suggestion bubbles (max 40 chars) + 1 “Subtext” tooltip (tap to expand).

Android Implementation Details:

- `InputMethodService` with `onStartInputView`.
- Fallback: if `InputConnection` fails, request Accessibility permission.

Data Flow:

1. Capture last 20 messages + current draft (JSON array).
2. Send via platform channel to main app (async, non-blocking).
3. Main app encrypts and calls `/analyze-intent`.
4. If latency > 800 ms: hide loading spinner and show “Thinking...” fallback.

3.3 Epic 2: Contextual Decoding & Scoring (Mathematical Model)

The scoring engine runs every 24h on the last 7 days of `analysis_logs`.

$$H_{\text{score}} = \left(w_1 \cdot \frac{C_{\text{direct}}}{C_{\text{total}}} \right) + \left(w_2 \cdot \sum_{i=1}^n M_i \right) - \left(w_3 \cdot \frac{R_{\text{escalation}}}{T_{\text{interactions}}} \right)$$

Where:

- C_{direct} : count of non-passive, explicit intent statements.
- C_{total} : total statements in window.
- M_i : mutually met emotional needs (0–1 per turn, NLP sentiment + need mapping).
- $R_{\text{escalation}}$: count of face-saving keywords (Cantonese list: “唔使理我”, “隨便啦”, “ok la”, English equivalents).
- $T_{\text{interactions}}$: conversation turns.
- Weights $w_1 = 0.4, w_2 = 0.35, w_3 = 0.25$ adjusted $\pm 10\%$ based on user baseline quiz.

Gemini System Prompt:

```
1 "You are RelateOS, a Hong Kong relationship coach. Analyze the following chat
   in context of Cantonese/English code-switching and face-saving culture.
   Never translate slang literally. Output only JSON with keys:
   subtext_explanation, suggestions (array of 3 objects with tone and text
   <=40 chars), health_delta, detected_emotion."
```

3.4 Epic 3: Wisdom Circle (Community Engine)

- Anonymous posting of redacted screenshots/chats.
- Edge Function (Deno) runs NER + rule-based redaction before insert.
- Redaction rules:
 - Replace any name, phone, HKID, email with [REDACTED].
 - Replace MTR stations with generic tags (e.g., “Causeway Bay” → “[Commercial District]”).
 - Replace exact addresses with “[Location]”.
- Upvote decay: visibility score = `upvote_count` $\times e^{-t/48}$ (t in hours).
- Moderation: auto-flag if Gemini confidence of toxicity > 0.7 .

Data Model (Supabase PostgreSQL)

Table	Column	Type / Constraints
users	id	uuid PRIMARY KEY DEFAULT uuid_generate_v4()
	created_at	timestampz DEFAULT now()
	preferred_language	varchar(10) CHECK (value IN ('en','zh-HK','zh-CN'))
	subscription_tier	varchar(20) DEFAULT 'free'
	onboarding_completed	boolean DEFAULT false
	consent_keyboard_granted	timestampz
	baseline_weights	jsonb

Table	Column	Type / Constraints
analysis_logs	id	uuid PK
	user_id	uuid REFERENCES users(id) ON DELETE CASCADE
	health_score_snapshot	float CHECK ($0 \leq value \leq 1$)
	primary_emotion_detected	varchar(50)
	created_at	timestampz
	raw_text_hash	varchar(64) (SHA-256 of redacted text only)

API Contracts

5.1 POST /api/v1/analyze-intent

```
1 Request: {  
2   "user_id": "uuid",  
3   "context_messages": ["msg1", "msg2", ...],  
4   "target_language": "zh-HK",  
5   "current_draft": "optional string"  
6 }  
7 Response: {  
8   "subtext_explanation": "string",  
9   "suggestions": [{ "tone": "string", "text": "string <=40 chars" }, ...],  
10  "health_delta": 0.5,  
11  "latency_ms": 420  
12 }
```

Non-Functional Requirements & Compliance

6.1 Privacy & Security

- Raw chat text NEVER persisted. Only SHA-256 hash stored.
- Payloads encrypted end-to-end (AES-256-GCM).
- Consent logged; revocation deletes data within 72 h.

6.2 Performance Budgets

- Keyboard render TTI: ≤ 300 ms.
- End-to-end AI roundtrip: ≤ 1200 ms on 4G.
- App cold start: ≤ 1.8 s.

Appendix A: Edge Case Matrix

(Omitted for brevity, but full prompts and 25 test cases are maintained in the central GitHub repository.)